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# -------------------------------------------------------
# Cell 1 : Import Libraries
# -------------------------------------------------------

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix


# -------------------------------------------------------
# Cell 2 : Load Dataset
# -------------------------------------------------------

df = pd.read_csv("Social_Network_Ads.csv")

print("Dataset Loaded Successfully\n")

print(df.head())


# -------------------------------------------------------
# Cell 3 : Dataset Information
# -------------------------------------------------------

print("\nDataset Information\n")
print(df.info())

print("\nStatistical Summary\n")
print(df.describe())


# -------------------------------------------------------
# Cell 4 : Check Column Names
# -------------------------------------------------------

print("\nColumn Names\n")
print(df.columns)
```

```
# -------------------------------------------------------
# Cell 5 : Select Features and Target
# -------------------------------------------------------

X = df[['Age', 'EstimatedSalary']]

y = df['Purchased']

print("\nTarget Values\n")
print(y.unique())


# -------------------------------------------------------
# Cell 6 : Train-Test Split
# -------------------------------------------------------

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42
)


# -------------------------------------------------------
# Cell 7 : Feature Scaling
# -------------------------------------------------------

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)


# -------------------------------------------------------
# Cell 8 : Train Logistic Regression Model
# -------------------------------------------------------

model = LogisticRegression()

model.fit(X_train, y_train)

print("\nModel Trained Successfully")
```

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# --------------------------------------------------------
# Cell 9 : Predictions
# --------------------------------------------------------

y_pred = model.predict(X_test)

print("\nPredicted Values\n")

print(y_pred[:10])


# --------------------------------------------------------
# Cell 10 : Confusion Matrix
# --------------------------------------------------------

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix\n")

print(cm)


# --------------------------------------------------------
# Cell 11 : TP, TN, FP, FN
# --------------------------------------------------------

TN = cm[0][0]

FP = cm[0][1]

FN = cm[1][0]

TP = cm[1][1]

print("\nTrue Positive (TP) :", TP)

print("True Negative (TN) :", TN)

print("False Positive (FP):", FP)

print("False Negative (FN):", FN)
```

```
# --------------------------------------------------------
# Cell 12 : Accuracy, Error Rate, Precision, Recall
# --------------------------------------------------------

accuracy = (TP + TN) / (TP + TN + FP + FN)

error_rate = 1 - accuracy

precision = TP / (TP + FP)

recall = TP / (TP + FN)

print("\nAccuracy    :", accuracy)

print("Error Rate :", error_rate)

print("Precision  :", precision)

print("Recall     :", recall)


# --------------------------------------------------------
# Cell 13 : Final Output
# --------------------------------------------------------

print("\nFinal Dataset Preview\n")

print(df.head())
```